COM-480 Milestone 2 - DataRizz

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1 Project Goal

The goal of our project is to provide insights into how inequalities have been reflected in the Olympic Games over time. We explore aspects such as gender, GDP, disciplines, geopolitics, and sociocultural contexts through our visualisations. We aim to present them through an interactive and user-friendly website.

2 Outline

Below, we present our proposed structure and visualisations for our website.

Where Do the Medal Winners Come From?

Understanding how Olympic medals are distributed across disciplines and regions can provide insight into global performance patterns and how they have evolved over time. The first section shows the distribution of gold, silver and bronze medals across the top 10 disciplines. Users can pick the plot between Summer and Winter editions, by default on Summer (Figure 1). The visualisation shows a Sankey diagram that illustrates the relative contribution of each continent. Users can select or deselect continents to display the corresponding countries for a given year. A time slider below the figure allows users to explore how this distribution has changed over time.

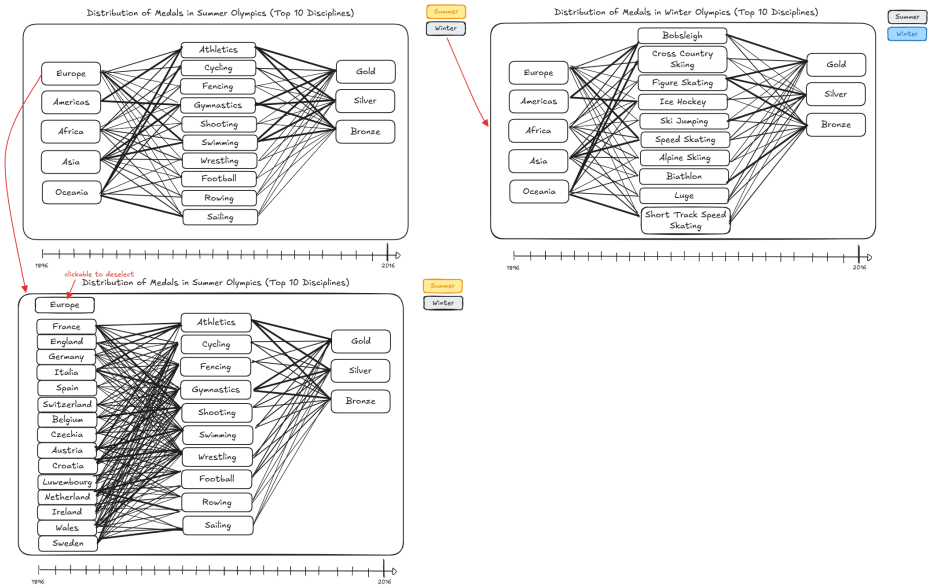

Figure 1: Medals around the world

How Has Gender Representation Evolved Across Olympic Sports and Disciplines Over Time?

Women first joined the Olympic Games in their second edition. However, it has taken more than a century to approach gender parity. To address this research question, Figure 2 represents the various sports and their disciplines using a sunburst plot for both men and women. The colour scale is lighter when the proportion of athletes of a given gender is lower and darker when it is higher. This visualisation highlights the evolution of the proportion of men and women in each discipline over the years. A time slider below the visualisation allows users to explore different years and can be paused at any moment. Users can also choose between Summer and Winter Olympic Games. When hovering over a sport or discipline, the exact percentage is displayed. In addition, we include a pie chart that illustrates the overall proportion of men and women across all disciplines.

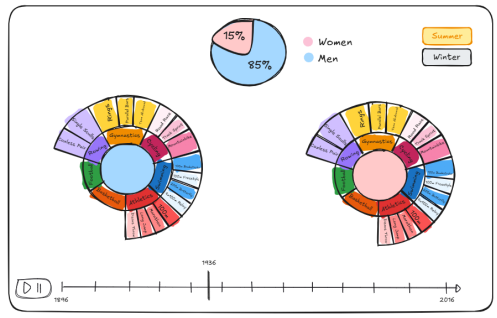

Figure 2: Gender evolution with pie chart and sunbursts

How Do Geopolitical Conflicts Affect the Olympic Games?

Major geopolitical conflicts, such as the World Wars, the Cold War, and Olympic boycotts, have shaped participation and performance in the Games. To address this question, users select a year, after which countries involved in a conflict are highlighted on a map. On the right side, we present key geopolitical events to guide the user. For example, the 1976 Montreal boycott led to 28 countries, mostly African, withdraw in protest against the participation of New Zealand, which was perceived as supporting apartheid. When users slide to the right, two additional visualisations (Figure 4) present some statistics. The left one shows the number of medals won by each country involved over time. The right one illustrates the number of participating countries. The time window is approximately ± 16 years around the selected year to see before, during and after evolution.

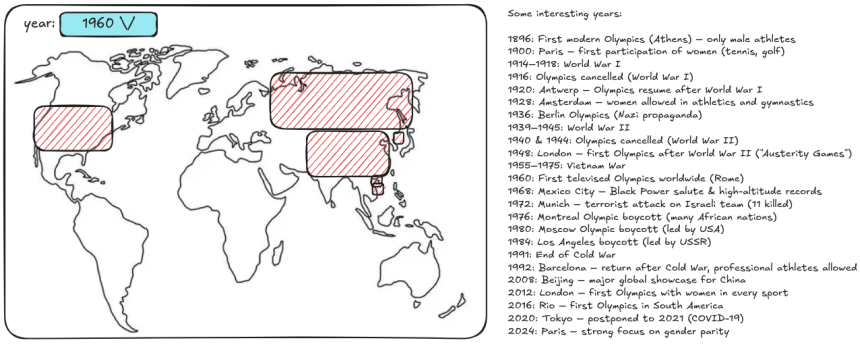

Figure 3: Geopolitical conflicts map

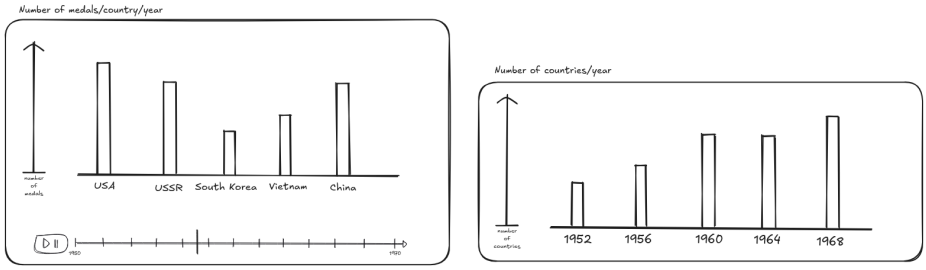

Figure 4: Geopolitical conflicts stats

What Would a Fair Distribution of Olympic Medals Look Like?

The geopolitical and sociocultural contexts affect the resources available to delegations at each Olympic Games, in terms of population size and GDP. These inequalities influence the outcome of the competitions. Out of curiosity, what would a fair distribution of Olympic medals look like? We compare the countries by scaling them to the average population and GDP. Through an interactive visualisation (Figure 5), we demonstrate this by applying course concepts such as filtering by season, year and country, aggregation using the fair medal plot and linked views for comparison. Users can pick the season (summer or winter), the year and two countries to be compared. We calculate the fair number of gold, silver and bronze medals as follows:

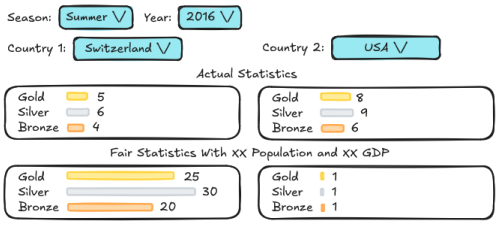

Figure 5: Fair medal distribution comparison

$$\#\text{ Fair medals} = \#\text{ Actual medals} \times \frac{\text{Mean population}}{\text{Country's population}} \times \frac{\text{Mean GDP}}{\text{Country's GDP}}$$

3 Tools

Our interactive website will be built using standard web technologies introduced throughout the course. We use GitHub Pages to host our website, allowing us to deploy and share it. For the skeleton version, we used `Fullpage.js` as a basis for a full screen one page scrolling website. We also used two Google Fonts: Carter One for headers and Delius for body. We used Excalidraw to create the sketches for this report. HTML and CSS will be used for structure and styling.

The above mentioned languages were introduced in Week 1 and will enable us to create clear, user-friendly layouts for our visualisations. JavaScript will be used to implement interactivity and dynamic content (Weeks 2 and 3). It will handle user inputs, updates to the visualisations, and coordination between the interface's different components. `D3.js` will be our main visualisation library (Weeks 4, 5 and others). It will be used to create customisable, interactive visualisations such as bar charts that evolve over time and zoomable sunburst diagrams. The user interactions will be designed based on the concepts from Week 5 (e.g., filtering, zooming and comparisons). To ensure readability and effective communication, we will follow the tips from Week 6 on colour perception for our visual encoding choices. Additionally, we will rely on the material from Week 8 on map design to implement the geographic visualisation. Furthermore, we will use content from Week 12 to make the website engaging, visually appealing and easy to navigate.

4 Additional Ideas

If time permits and the ideas are feasible, we would like to include the following features in our final website:

1. A first idea would be to create an interactive visualisation showing the body types of the medallists. Users could filter by season, year, gender, sport and discipline. As in Figure 6, it would display a podium with the body types of the winners, indicating their height, weight, name and delegation. To implement this, we could render the scene in 3D using `Three.js`, load a human model (MakeHuman), apply morph targets to adjust weight and supposed BMI (Blender), and scale the model to the correct height.

2. Another idea would be to add interesting facts to the "Where Do the Medal Winners Come From?" visualisation to help explain some of the trends observed in the data. These facts would provide additional context and make the visualisation easier to understand. For each year, one or two key insights explaining the width of certain flows in the Sankey diagram would be highlighted and presented below the plot, allowing users to better interpret the data. The visualisation in Figure 7 represents this idea.

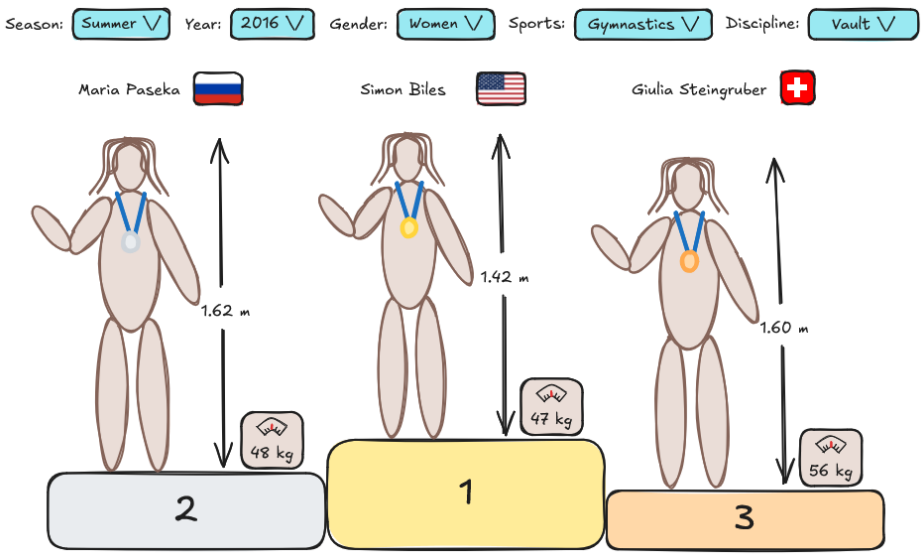

Figure 6: Podium of body types

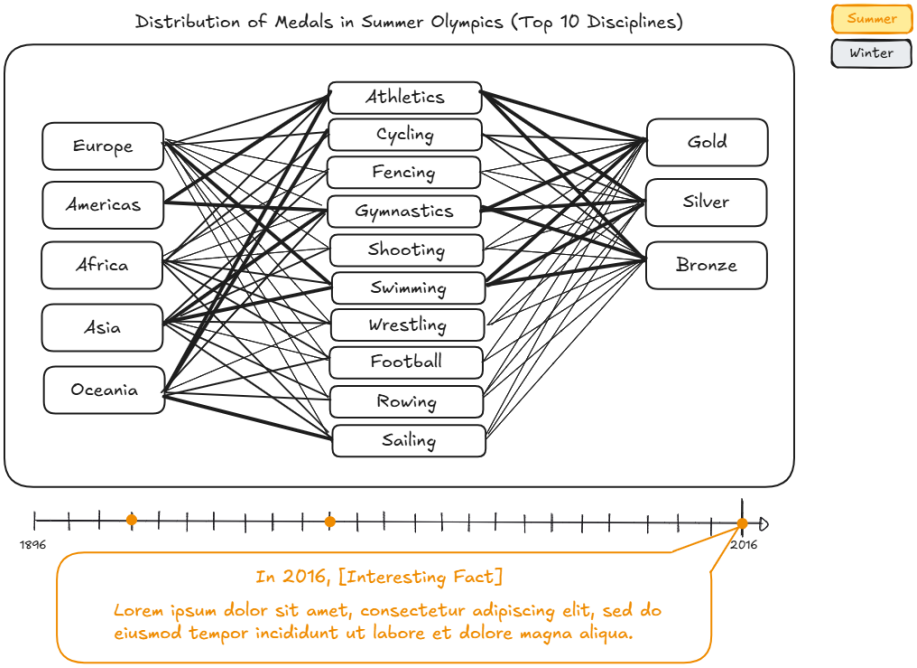

Figure 7: First visualisation with interesting facts